

## Function Stack Tracing Practice

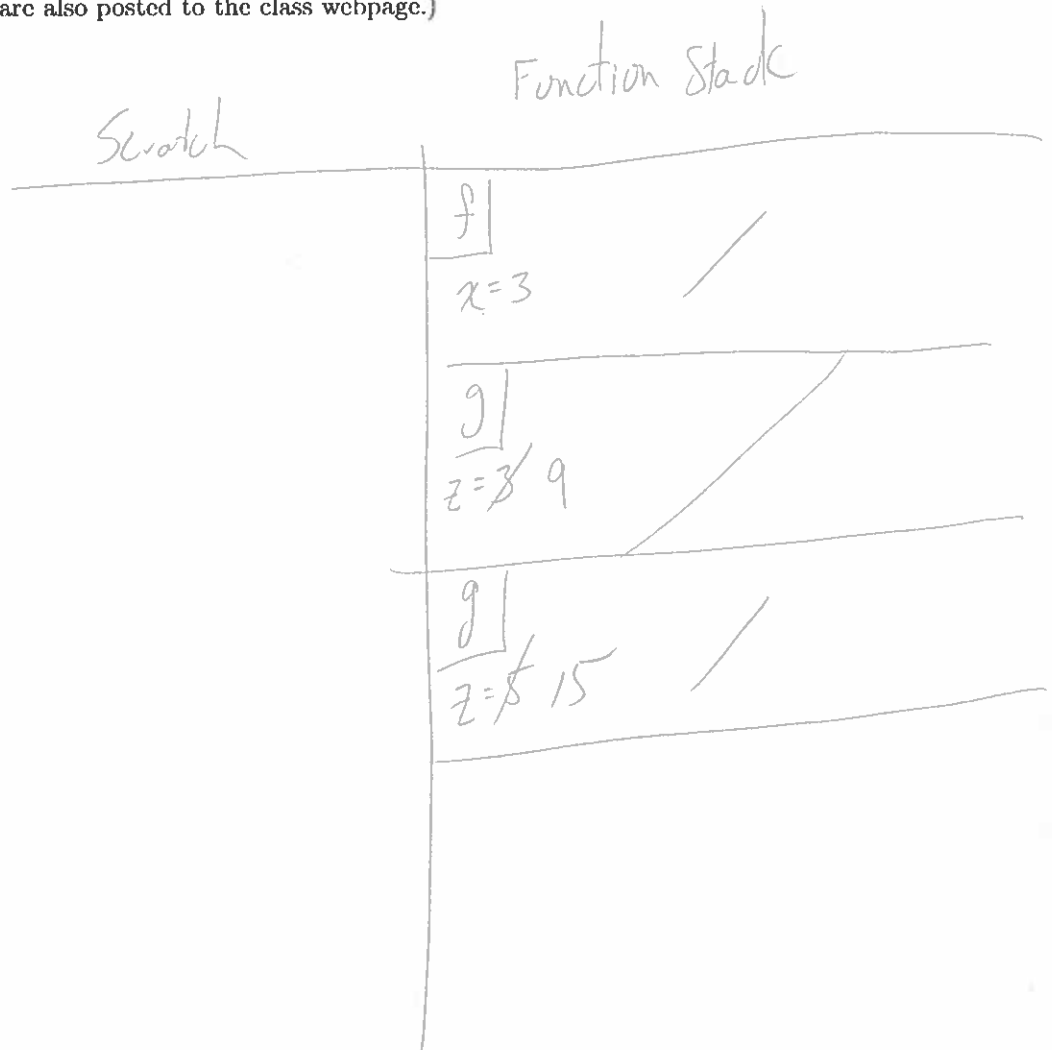
(This code, and sample solutions, are also posted to the class webpage.)

1.

```
def g(z: int):  
    z = z * 3  
    print(z)
```

```
def f(x: int):  
    g(x)  
    g(x + 2)  
    print(x)
```

```
f(3)
```



Printed Output

9  
15  
3

Note: A new copy of g() goes on the stack each time it is called!

2.

```
def alice(z : int) -> int:
    if z > 0:
        return z * 2
    else:
        return z + 10

def bob(x : int, y : int) -> int:
    if y < x:
        return alice(x) + alice(y)
    else:
        return alice(x + y)

def main2():
    print(bob(6,7))

main2()
```

## Function Stack

main2()

bob

x = 6  
y = 7

26

alice

z = 13

26

Returned  
Values  
So here

Printed output

26

[We then redo this example, but change the code inside main]

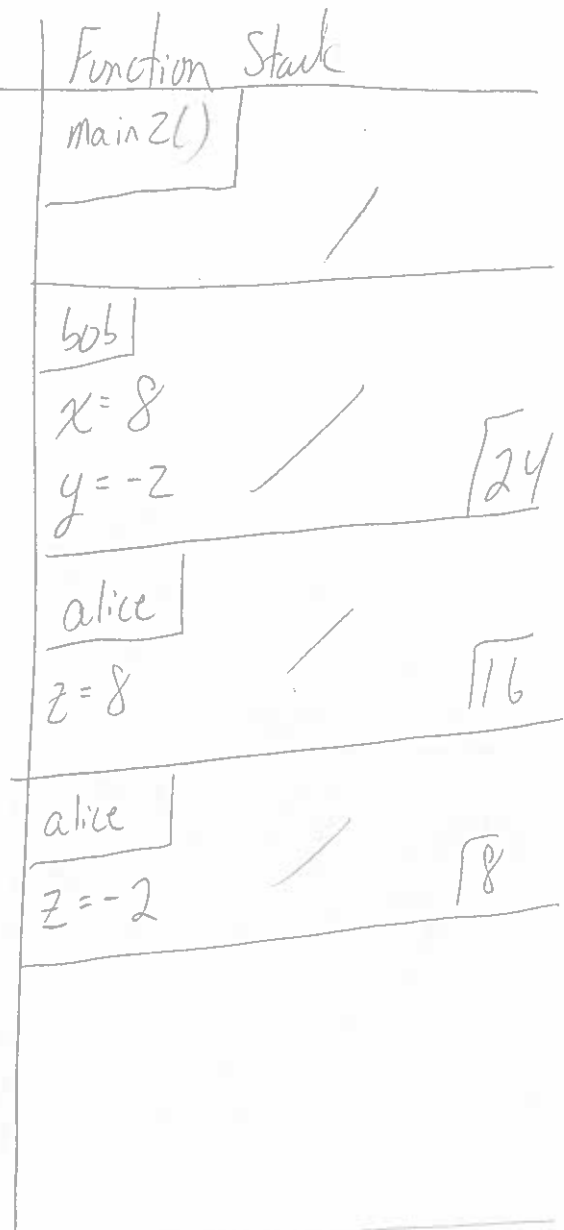
2.5.

```
def alice(z : int) -> int:
    if z > 0:
        return z * 2
    else:
        return z + 10

def bob(x : int, y : int) -> int:
    if y < x:
        return alice(x) + alice(y)
    else:
        return alice(x + y)

def main2():
    print(bob(8, -2))

main2()
```



Printed output

24

Notice alice  
is called twice.

3.

```
def aaa(x: int, s: str) -> int:
    z = 7
    if x < z and s < 'hello':
        z += 3
        print(f'My string is {s} and number is {x}.')
    else:
        z *= 4
        print(f'I do not like the number {x}.')
    return z
```

```
def bbb(x: int, y: int) -> int:
    if x < y:
        return y // x
    else:
        return x - y
```

```
def main3():
    x = bbb(7,2)
    print(aaa(x,'hi'))
    print(aaa(bbb(3,14),'abcd'))
```

```
main3()
```

Function Stack

main3

x = 5

bbb

x = 7

y = 2

5

aaa

x = 5

s = 'hi'

z = 28

28

aaa

x = 4

s = 'abcd'

z = 10

10

bbb

x = 3

y = 14

4

I do not like the number 5.

28

My string is abcd and number is 4.

10